

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-I]

TEST # 02

DATE : 13-11-2019

REVISION TEST

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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REVISION TEST # 02

Topics

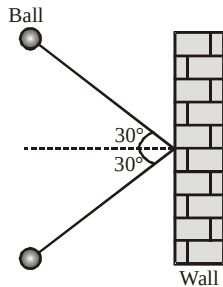
DATE : 13 - 11 - 2019**PRE-MEDICAL ENTHUSIAST COURSE : PHASE - I – PATTERN : [NEET]****PHYSICS** : Relative Motion, NLM (Basic)**CHEMISTRY** : Chemical Bonding**BOTANY** : Cell Biology, Mendelian Genetics (Upto Mono & Dihybrid cross)

SPACE FOR ROUGH WORK

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

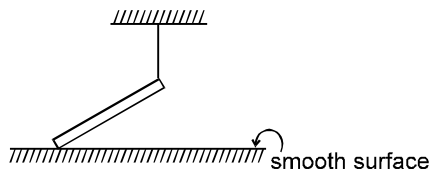
BEWARE OF NEGATIVE MARKING

1. A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 30° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the average force exerted by the ball on the wall is :-



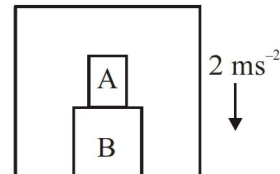
- (1) $1500\sqrt{3}$ N (2) 1500 N
(3) $300\sqrt{3}$ N (4) 300 N

2. Which figure represents the correct F.B.D. of rod of mass m as shown in figure :



- (1) (2)
(3) (4) None of these

3. The elevator shown in figure is descending with acceleration of 2 ms^{-2} . The mass of the block A is 0.5 kg. The force exerted by block A on the block B is :-



- (1) 2 N (2) 4 N
(3) 6 N (4) 8 N

4. An elevator in which a man is standing is moving upwards with a speed of 10 m/sec. If a man drops a coin from a height of 2.45 m, it reaches the floor of the elevator after a time :-

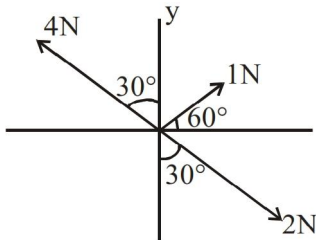
- (1) $\sqrt{2}$ sec (2) $1/\sqrt{2}$ sec
(3) 2 sec (4) $\frac{1}{2}$ sec

5. A motor boat covers a given distance in 6 hours moving down stream of a river. It covers the same distance in 10 hours moving upstream. The time (in hours) it takes to cover the same distance in still water is :

- (1) 6 (2) 7.5
(3) 10 (4) 15

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6. Three forces acting on a body are shown in figure. To have the resultant force only along y-direction, the magnitude of the minimum additional force needed is :-



- (1) 0.5 N (2) 1.5 N
(3) $\frac{\sqrt{3}}{4}$ N (4) $\sqrt{3}$ N
7. If $\vec{a}_1$ and $\vec{a}_2$ are two non collinear unit vectors and if $|\vec{a}_1 + \vec{a}_2| = \sqrt{3}$, then the value of $(\vec{a}_1 - \vec{a}_2) \cdot (2\vec{a}_1 + \vec{a}_2)$ is :-
- (1) 2 (2) $\frac{3}{2}$ (3) $\frac{1}{2}$ (4) 1
8. A thief is running away on a straight road in a jeep moving with a speed of 9 ms^{-1} . A policeman chases him on a motor cycle moving at a speed of 10 ms^{-1} . If the instantaneous separation of the jeep from the motor cycle is 100 m, how long will it take for the policeman to catch the thief?
- (1) 1 s (2) 19 s
(3) 90 s (4) 100 s
9. If R is the maximum horizontal range of a particle, then the greatest height attained by it is :-
- (1) R (2) 2R (3) $\frac{R}{2}$ (4) $\frac{R}{4}$

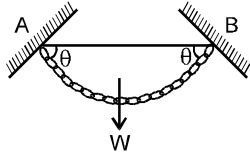
10. During a projectile motion if the maximum height equals to the horizontal range, then the angle of projection with the horizontal is :-
- (1) $\tan^{-1}(1)$ (2) $\tan^{-1}(2)$
(3) $\tan^{-1}(3)$ (4) $\tan^{-1}(4)$
11. The point from where a ball is projected is taken as the origin of the co-ordinate axes. The x and y components of its displacement are given by $x = 6t$ and $y = 8t - 5t^2$. What is the velocity of projection?
- (1) 6 ms^{-1} (2) 8 ms^{-1}
(3) 10 ms^{-1} (4) 14 ms^{-1}
12. A particle is fired with velocity u making an angle θ with the horizontal. What is the change in velocity when it is at the highest point?
- (1) $u \cos \theta$ (2) u
(3) $u \sin \theta$ (4) $(u \cos \theta - u)$
13. A ball whose kinetic energy is E, is thrown at an angle of 45° with the horizontal, its kinetic energy at the highest point of its flight will be :-
- (1) E (2) $\frac{E}{\sqrt{2}}$
(3) $\frac{E}{2}$ (4) zero
14. A bullet is dropped from the same height when another bullet is fired horizontally. They will hit the ground :-
- (1) simultaneously
(2) depends on the observer
(3) one after the other
(4) none of these

SPACE FOR ROUGH WORK

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| <p>15. An aeroplane moving horizontally with a speed of 180 km/hr drops a food packet while flying at a height of 490 m. The horizontal range is :-
 (1) 180 m (2) 980 m
 (3) 500 m (4) 670 m</p> <p>16. A train of 150 m length is going towards north direction at a speed of 10 ms^{-1}. A parrot flies at a speed of 5 ms^{-1} towards south direction parallel to the railway track. The time taken by the parrot to cross the train is equal to :-
 (1) 12 s (2) 8 s
 (3) 15 s (4) 10 s</p> <p>17. A train 100 m long travelling at 40 ms^{-1} overtakes another train 200 m long travelling at 30 ms^{-1}. The time taken by the first train to pass the second train is :-
 (1) 30 s (2) 40 s (3) 50 s (4) 60 s</p> <p>18. A car starts from rest and moves with constant acceleration. The ratio of the distance covered in the n^{th} second to that covered in n seconds is :-
 (1) $\frac{2}{n^2} - \frac{1}{n}$ (2) $\frac{2}{n^2} + \frac{1}{n}$
 (3) $\frac{2}{n} - \frac{1}{n^2}$ (4) $\frac{2}{n} + \frac{1}{n^2}$</p> <p>19. A particle's position as a function of time is described as $y(t) = 2t^2 + 3t + 4$. What is the average velocity of the particle from $t = 0$ to $t = 3 \text{ sec}$?
 (1) 3 m/s (2) 6 m/s
 (3) 9 m/s (4) 12 m/s</p> | <p>20. An object of mass 10 kg moves at a constant speed of 10 m/s. A constant force, that acts for 4 s on the object, gives it a speed 2 m/s in opposite direction. The force acting on the object is :-
 (1) -3 N (2) -30 N
 (3) 3 N (4) 30 N</p> <p>21. A person standing on an escalator takes time t_1 to reach the top of a tower when the escalator is moving. He takes time t_2 to reach the top of the tower when the escalator is standing. How long will he take if he walks up a moving escalator?
 (1) $t_2 - t_1$
 (2) $t_1 + t_2$
 (3) $t_1 t_2 / (t_1 - t_2)$
 (4) $t_1 t_2 / (t_1 + t_2)$</p> <p>22. A bus starts moving with acceleration 2 ms^{-2}. A cyclist 96 m behind the bus starts simultaneously towards the bus at 20 m/s. After what time will he be able to overtake the bus?
 (1) 4 s (2) 8 s
 (3) 12 s (4) 16 s</p> <p>23. Two cars are moving in the same direction with the same speed of 30 km/h. They are separated by 5 km. What is the speed of the car moving in the opposite direction if it meets the two cars at an interval of 4 minutes?
 (1) 15 km/h
 (2) 30 km/h
 (3) 45 km/h
 (4) 60 km/h</p> |
|---|--|

SPACE FOR ROUGH WORK

24. A flexible chain of weight W hangs between two fixed points A and B at the same level. The inclination of the chain with the horizontal at the two points of support is θ . What is the tension of the chain at the endpoint.



- (1) $\frac{W}{2} \operatorname{cosec} \theta$ (2) $\frac{W}{2} \sec \theta$
(3) $W \cos \theta$ (4) $\frac{W}{3} \sin \theta$

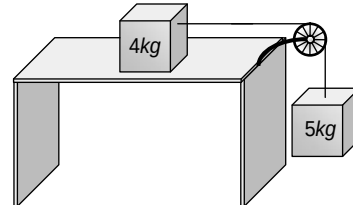
25. From the top of the tower of height 400 m, a ball is dropped by a man, simultaneously from the base of the tower, another ball is thrown up with velocity 50 m/s; at what distance will they meet from the base of the tower?

- (1) 100 m (2) 320 m
(3) 80 m (4) 240 m

26. A body A starts from rest with an acceleration a_1 . After 2 seconds, another body B starts from rest with an acceleration a_2 . If they travel equal distances in the 5th second, after the start of A, then the ratio $a_1 : a_2$ is equal to :-

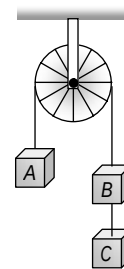
- (1) 5 : 9 (2) 5 : 7
(3) 9 : 5 (4) 9 : 7

27. Two masses of 4 kg and 5 kg are connected by a string passing through a frictionless pulley and are kept on a frictionless table as shown in the figure. The acceleration of 5 kg mass is



- (1) 49 m/s² (2) 5.44 m/s²
(3) 19.5 m/s² (4) 2.27 m/s²

28. Three equal weights A, B and C of mass 2 kg each are hanging on a string passing over a fixed frictionless pulley as shown in the figure. The tension in the string connecting weights B and C is



- (1) Zero (2) 13 N
(3) 3.3 N (4) 19.6 N

29. A mass 1 kg is suspended by a thread. It is

- (i) lifted up with an acceleration 4.9 m/s²
(ii) lowered with an acceleration 4.9 m/s².

The ratio of the tensions is :

- (1) 3 : 1 (2) 1 : 3
(3) 1 : 2 (4) 2 : 1

SPACE FOR ROUGH WORK

30. Force $F = 100 \text{ N}$ is applied on a combination of mass M and m in two situations as shown in figure (i) & (ii). If contact force between M & m in two cases is N_1 & N_2 respectively

then calculate $\frac{N_2}{N_1}$.

(Given : $M = 10 \text{ kg}$, $m = 2 \text{ kg}$). Friction is absent everywhere.

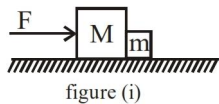


figure (i)

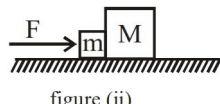
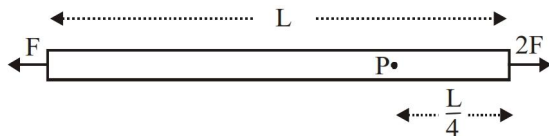
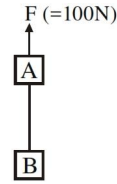


figure (ii)

- (1) 2 (2) 5
(3) 4 (4) 6
31. A uniform rod of mass M and length L lies flat on a frictionless horizontal surface. Two forces F and $2F$ are applied along the length of the rod as shown. The tension in the rod at point P is



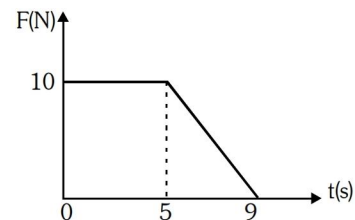
- (1) F (2) $3F$
(3) $\frac{5F}{4}$ (4) $\frac{7F}{4}$
32. Consider the shown arrangement where the blocks A and B connected by means of a uniform string is being moved vertically up by the force F . Each block weighs 2 kg while the mass of string is 1000 gm . The tension at bottom of the string equals



- (1) 20 N (2) 40 N
(3) 60 N (4) 270 N
33. A uniform rope of mass 1.0 kg is connected with a box of mass 2.0 kg , which is placed on a smooth horizontal surface. The free end of the rope is pulled horizontally by a force 6 N . Find the tension at the midpoint of the rope.



- (1) 4 N (2) 1 N
(3) 5 N (4) 2 N
34. A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is :-



- (1) 280 N-s (2) 140 N-s
(3) 70 N-s (4) 210 N-s
35. A bird weighs 2 kg and is inside a closed cage of 1 kg . If it starts flying with constant velocity, then what is the weight of the bird and cage assembly
- (1) 1.5 kg (2) 2.5 kg
(3) 3 kg (4) 4 kg

SPACE FOR ROUGH WORK

36. Which of the following statement is correct :
- (1) If force is along velocity, then it can change the direction of motion.
 - (2) If force is perpendicular to velocity, then it can change the magnitude of velocity,
 - (3) If force is in the opposite direction of velocity, then it can not change the direction of motion.
 - (4) If force is perpendicular to velocity then it can change direction of velocity.
37. A force ' F_0 ' start acting on a stationary particle of mass ' m ' then velocity of particle after covering a distance ' d ' is :-
- (1) $\sqrt{\frac{m}{2F_0d}}$
 - (2) $\frac{2F_0d}{m}$
 - (3) $\sqrt{\frac{2F_0d}{m}}$
 - (4) Zero
38. A bullet of mass 0.08 kg moving with a speed of 80m/s enters a heavy wooden block and is stopped after a distance of 40cm. What will be average resistive force on bullet by block.
- (1) 800 N
 - (2) 400 N
 - (3) 640 N
 - (4) 720 N
39. A bullet of mass 5 gram is accelerated in a rifle barrel with an approximately constant force of 2500 N. The mass of the rifle is 5 kg. What is the force pushing the rifle back ?
- (1) 2.5 N
 - (2) 2500 N
 - (3) 250000 N
 - (4) 0 N
40. A particle of mass 2kg moves in free space such that its position vector varies with time as $\vec{r} = [(3+4t^2)\hat{i} + (2t)\hat{j} + (3-6t)\hat{k}]m$ where t is in second. Net force acting on the particle is
- (1) zero
 - (2) parallel to x-axis
 - (3) parallel to y-axis
 - (4) time dependent
41. A machine gun fires 10 bullets per second in horizontal direction. Mass of each bullet is 80 g and muzzle velocity is 500 m/s. If the average horizontal force the shooter must apply to keep the machine gun stationary is $F \times 10^2$ N, find numerical value of F .
- (1) 4
 - (2) 5
 - (3) 8
 - (4) 12

SPACE FOR ROUGH WORK

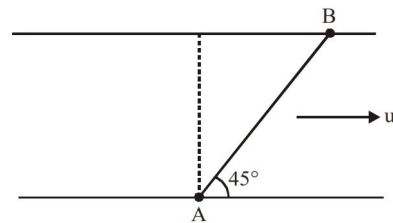
42. A cricket ball of mass 160 g is moving horizontally directly towards a batsman. Its speed just before it hits the bat is 30 m/s. It leaves the bat at 40 m/s at 90° to its original direction. Find the magnitude of the impulse (in N-s) imparted to the ball.

- (1) 4.8 (2) 8
(3) 6.4 (4) 10

43. A swimmer who can swim in a river with speed αv (with respect to still water) where v is the velocity of river current, jumps into the river from one bank to cross the river. Then choose wrong statement.

- (1) If $\alpha < 1$ he can not cross the river
(2) If $\alpha \leq 1$ he can not reach a point on other bank directly opposite to his starting point
(3) If $\alpha > 1$ he can reach a point on other bank directly opposite to his starting point
(4) He can reach the other bank at some point, for any non zero value of α .

44. A man wants to reach point B on the opposite, bank of a river flowing at a speed 'u' as shown in the figure. What minimum speed relative to water should the man have so that he can reach directly to the point B ?



- (1) $\frac{u}{\sqrt{2}}$, 30° with upstream
(2) $\frac{u}{\sqrt{2}}$, 45° with downstream
(3) $\frac{u}{\sqrt{2}}$, 45° with upstream
(4) $\frac{u}{\sqrt{2}}$, 30° with downstream

45. Rain is falling with a velocity

$(-4\hat{i} + 8\hat{j} - 10\hat{k})$ m/s. A person is moving with a velocity of $(6\hat{i} + 8\hat{j})$ m/s on the ground. The speed with which the rain drops hit the person is :-

- (1) 10 m/s
(2) $10\sqrt{2}$ m/s
(3) $\sqrt{180}$ m/s
(4) $\sqrt{360}$ m/s

SPACE FOR ROUGH WORK

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| <p>46. A σ-bonded molecule MX_3 is T-shaped. The number of lone pairs of electrons can be
 (1) 0 (2) 2
 (3) 1 (4) none of these</p> <p>47. Which of the following should have pyramidal shape :
 (1) $[ClOF_2]^+$ (2) ICl_3
 (3) $[BrICl]^-$ (4) All of these</p> <p>48. Which of the following molecules has two lone pairs and bond angle (need not be all bond angles) $< 109.5^\circ$?
 (1) SF_2 (2) KrF_4
 (3) ICl_4^- (4) All of these</p> <p>49. The correct order of bond angle is :
 (1) $H_2S < NH_3 < BF_3 < CH_4$
 (2) $NH_3 < H_2S < CH_4 < BF_3$
 (3) $H_2S < NH_3 < CH_4 < BF_3$
 (4) $H_2S < CH_4 < NH_3 < BF_3$</p> <p>50. In which of the following molecules are all the bonds not equal?
 (1) NF_3 (2) ClF_3
 (3) BF_3 (4) AlF_3</p> <p>51. Which of the following is correct order of bond length ?
 (1) $BF_4^- < BF_3$ (2) $NO_2^+ < NO_2^-$
 (3) $CCl_4 < CF_4$ (4) $^+CH_3 > CH_4$</p> <p>52. Identify the correct statement:
 (1) single N-N bond is stronger than single P-P bond
 (2) single N-N bond is weaker than single P-P bond
 (3) $N \equiv N$ is weaker than $P \equiv P$
 (4) None of these</p> | <p>53. In which of the following species peroxide group is not present :
 (1) $[B_4O_5(OH)_4]^{2-}$ (2) $[S_2O_8]^{2-}$
 (3) CrO_5 (4) HNO_4</p> <p>54. Which of the following is correct ?
 (1) S_3O_9: contains no S-S linkage.
 (2) $S_2O_6^{2-}$: contains -O-O- linkage.
 (3) $(HPO_3)_3$: contains P-P linkage
 (4) $S_2O_8^{2-}$: contains S-S linkage</p> <p>55. The percentage of s-character in the orbital forming P-S bonds in P_4S_3 is :
 (1) 25 (2) 33 (3) 75 (4) 50</p> <p>56. Indicate the incorrect statement:
 (1) Number of hybrid orbitals formed is equal to no. of atomic orbitals involved.
 (2) $2p_x$ and $2p_y$ - orbitals of carbon can be hybridized to yield two new more stable orbitals
 (3) Effective hybridisation is not possible with orbitals of widely different energies
 (4) The concept of hybridisation has a greater significance in the VB theory of localised orbitals than in the MO theory.</p> <p>57. In which of the following compounds B atoms are in sp^2 and sp^3 hybridisation states ?
 (1) Borax (2) Diborane
 (3) Borazole (4) All</p> |
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SPACE FOR ROUGH WORK

58. S₁: Oxidation number of N in N₂O₅ is 5
 S₂: The anhydride of Hypochlorous acid is Cl₂O
 S₃: Bond angle of NH₃ and NF₃ are same.
 S₄: For heteronuclear diatomic species A-B, the bond length decreases as the difference in electronegativity values increases (considering A and B of similar size) in all compound.
- (1) T T T F (2) F T T T
 (3) F F T F (4) T T F T
59. In which of the following compounds B-F bond length is shortest ?
 (1) BF₄⁻ (2) BF₃ → NH₃
 (3) BF₃ (4) BF₃ → N(CH₃)₃
60. Which of the following statement is false for trisilylamine ?
 (1) Three sp² orbitals are used for σ bonding, giving a plane triangular structure.
 (2) Three sp³ orbitals are used for σ bonding, giving a plane triangular structure.
 (3) The N-Si bond length is shorter than the expected N-Si bond length.
 (4) It is a weaker Lewis base than trimethyl amine.
61. In which of the following molecules/species all following characteristics are found ?
 (a) Tetrahedral hybridisation
 (b) Hybridisation can be considered to have taken place with the help of empty orbital(s).
 (c) All bond lengths are identical i.e. all A-B bond lengths are identical.
- (1) B₂H₆ (2) Al₂Cl₆
 (3) BeCl₂(g) (4) BF₄⁻
62. H-bonding is maximum in
 (1) C₆H₅OH
 (2) C₆H₅COOH
 (3) CH₃CH₂OH
 (4) CH₃COCH₃
63. Which one of the following have least intermolecular H-bonding ?
 (1) H₂O (2) o-nitro phenol
 (3) HF (4) CH₃COOH
64. Which of the following is/are correct.
 (a) Boiling point of alcohol is higher than than of diethyl ether
 (b) Density of water is higher than ice.
 (c) Glycerol is more viscous than ethanol
 (d) Ammonia is more easily liquified than HCl due to H-bonding in NH₃
- (1) a, b, c (2) b, c, d
 (3) a, d (4) a, b, c, d
65. Which of the following statements is correct regarding phosphoric acid ?
 (a) pπ-dπ back bonding exist between O and P.
 (b) The anion is resonance stablized.
 (c) It is a dibasic acid.
 (d) Inter molecular H bonding between molecules make it a syrupy (viscous) liquid.
- (1) a, b, d (2) b, d
 (3) a, b, c (4) a, b, c, d

SPACE FOR ROUGH WORK

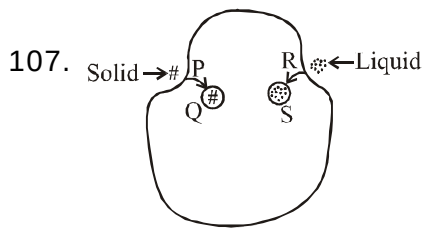
66. Which of the following is correct order of strength of hydrogen bonding?
 (a) $\text{N}-\text{H} \cdots \text{N} > \text{N}-\text{H} \cdots \text{O}$
 (b) $\text{F}-\text{H} \cdots \text{N} > \text{O}-\text{H} \cdots \text{N}$
 (c) $\text{N}-\text{H} \cdots \text{Cl} > \text{N}-\text{H} \cdots \text{N}$
 (d) $\text{O}-\text{H} \cdots \text{F} > \text{O}-\text{H} \cdots \text{O}$
 (1) a, b (2) a, b, c
 (3) c, d (4) a, b, c
67. The state of hybridisation of central atom in dimer form of both BH_3 and BeH_2 is
 (1) sp^2 , sp (2) sp^3 , sp^2
 (3) sp^3 , sp^3 (4) sp^2 , sp^3
68. Which of the following molecule has complete octet
 (1) B_2H_6 (2) Al_2Cl_6
 (3) Be_2Cl_4 (4) BeH_2
69. If Z-axis is the molecular axis, then π -molecular orbitals are formed by the overlap of
 (1) $s + p_z$ (2) $p_x + p_y$
 (3) $p_z + p_z$ (4) $p_x + p_x$
70. Bond order is a concept in the molecular orbital theory. It depends on the number of electrons in the bonding and antibonding orbitals. Which of the following statements is true about it? The bond order
 (1) Can have a negative quantity
 (2) Has always an integral value
 (3) Can assume any positive or integral or fractional value including zero
 (4) Is a non zero quantity
71. Which of the following pairs have identical values of bond order?
 (1) N_2^+ and O_2^+ (2) F_2 and Ne_2
 (3) O_2 and B_2 (4) C_2 and N_2
72. Which of the following molecules/ions exhibit sp mixing?
 (1) B_2 (2) C_2^{2-}
 (3) O_2^+ (4) Both (1) and (2)
73. The common features of the species N_2^{2-} , O_2 and NO^- are :
 (1) bond order three and isoelectronic.
 (2) bond order two and isoelectronic.
 (3) bond order three but not isoelectronic.
 (4) bond order two but not isoelectronic.
74. Which of the following molecular orbitals has two nodal planes.
 (1) $\sigma 2s$ (2) $\pi 2p_y$
 (3) $\pi^* 2p_y$ (4) $\sigma^* 2p_x$
75. Among the following species, which has the minimum bond length?
 (1) B_2 (2) C_2 (3) F_2 (4) O_2^-
76. Which of the following species is paramagnetic?
 (1) NO^- (2) O_2^{2-} (3) CN^- (4) CO
77. The following molecules / species have been arranged in the order of their increasing bond orders, Identify the correct order.
 (I) O_2 (II) O_2^- (III) O_2^{2-} (IV) O_2^+
 (1) III < II < I < IV (2) IV < III < II < I
 (3) III < II < IV < I (4) II < III < I < IV

SPACE FOR ROUGH WORK

78. Among the following which one will have the largest O – O bond length ?
 (1) KO_2 (2) O_2
 (3) $\text{O}_2^+[\text{AsF}_6]^-$ (4) K_2O_2
79. The correct order in which the O – O bond length increases in the following is :
 (1) $\text{H}_2\text{O}_2 < \text{O}_2 < \text{O}_3$ (2) $\text{O}_2 < \text{H}_2\text{O}_2 < \text{O}_3$
 (3) $\text{O}_2 < \text{O}_3 < \text{H}_2\text{O}_2$ (4) $\text{O}_3 < \text{H}_2\text{O}_2 < \text{O}_2$
80. Which of the following is a wrong order with respect to the property mentioned against each ?
 (1) $\text{O}_2^{2-} > \text{O}_2 > \text{O}_2^+$ [Paramagnetic moment]
 (2) $(\text{NO})^- > (\text{NO}) > (\text{NO})^+$ [bond length]
 (3) $\text{H}_2 > \text{H}_2^+ > \text{He}_2^+$ [bond energy]
 (4) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$ [bond angle]
81. Which of the following option with respect to increasing bond dissociation energies is correct ?
 (1) $\text{NO} < \text{C}_2 < \text{O}_2^- < \text{He}_2^+$
 (2) $\text{C}_2 < \text{NO} < \text{He}_2^+ < \text{O}_2^-$
 (3) $\text{He}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2$
 (4) $\text{He}_2^+ < \text{O}_2^- < \text{C}_2 < \text{NO}$
82. Pick out the incorrect statement.
 (1) N_2 has greater dissociation energy than N^{2+}
 (2) O_2 has lower dissociation energy than O_2^+
 (3) Bond length in N^{2+} is less than N_2
 (4) Bond length in NO^+ is less than in NO .
83. The species which are diamagnetic :
 (1) O^{2-} (2) NO_2
 (3) ClO_2 (4) N_2O_4
84. Which of the following is in order of increasing covalent character ?
 (1) $\text{CCl}_4 < \text{BeCl}_2 < \text{BCl}_3 < \text{LiCl}$
 (2) $\text{LiCl} < \text{CCl}_4 < \text{BeCl}_2 < \text{BCl}_3$
 (3) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 (4) $\text{LiCl} < \text{BeCl}_2 < \text{CCl}_4 < \text{BCl}_3$
85. Which of the following combination of ion will have highest polarisation ?
 (1) $\text{Fe}^{2+}, \text{Br}^-$ (2) $\text{Ni}^{4+}, \text{Br}^-$
 (3) $\text{Ni}^{2+}, \text{Br}^-$ (4) Fe, Br^-
86. The correct order of decreasing polarizability of ion is :
 (1) $\text{Cl}^-, \text{Br}^-, \text{I}^-, \text{F}^-$ (2) $\text{F}^-, \text{I}^-, \text{Br}^-, \text{Cl}^-$
 (3) $\text{I}^-, \text{Br}^-, \text{Cl}^-, \text{F}^-$ (4) $\text{F}^-, \text{Cl}^-, \text{Br}^-, \text{I}^-$
87. SnCl_4 is a covalent liquid because :
 (1) electron clouds of the $\text{Cl}^\ominus$ ions are weakly polarized to envelop the cation.
 (2) electron clouds of the $\text{Cl}^\ominus$ ions are strongly polarized to envelop the cation.
 (3) its molecules are attracted to one another by strong van der Waals forces.
 (4) Sn shows inert pair effect.
88. Which of the following compounds possesses zero dipole moment?
 (1) Benzene (C_6H_6)
 (2) Carbon tetrachloride
 (3) Boron trifluoride
 (4) All of these
89. Of the following molecules, the one, which has permanent dipole moment?
 (1) SiF_4 (2) CO_2
 (3) PF_3 (4) PF_5
90. The geometry of H_2S and its dipole moment are :-
 (1) angular and non zero
 (2) angular and zero
 (3) linear and non zero
 (4) linear and zero

SPACE FOR ROUGH WORK

91. Ribosomes in prokaryotes
(A) Are 70 S type
(B) Not bounded by any membrane
(C) They have 50 S and 30 S subunit
(D) Form polysomes
(1) (A), (B), (C) are correct
(2) Only (C) is correct
(3) All are correct
(4) (A), (B) and (D) are correct
92. Blue and red colour of petals is mainly due to :-
(1) Anthocyanin
(2) Fucoxanthin
(3) Phycoerythrin
(4) Haemoglobin
93. Which of the following plastid store protein?
(1) Aleuroplast (2) Amyloplast
(3) Chromoplast (4) Elaioplast
94. Mitochondria and chloroplasts are :-
(1) Considered as part of the endomembrane system
(2) Single membrane bound cell organelles
(3) Divide by binary fission
(4) Having 80 S ribosomes
95. Ribosomes of mitochondria, chloroplast and R.E.R are respectively :-
(1) 80S, 70S, 70S
(2) 80S, 80S, 80S
(3) 70S, 70S, 80S
(4) 70S, 80S, 70S
96. Which of the following has direct connections to outer membrane of nuclear membrane?
(1) RER (2) Golgi bodies
(3) Lysosomes (4) Mitochondria
97. Plasma membrane is regarded as asymmetric because :-
(1) Carbohydrate in outer and spectrin protein inner sides are present
(2) It contains carbohydrate
(3) Protein icebergs in sea of lipid
(4) Made up of glycoprotein and lipid
98. Choose the incorrect match
(1) Cristae – Infolding of inner membrane of mitochondria
(2) Granna – Stacks of thylakoids
(3) Ribosome – Protein synthesis
(4) Elaioplasts – Store protein granules
99. In human being, the RBC membrane has approximately :-
(1) 52% protein, 40% lipid
(2) 40% protein, 52% lipid
(3) 42% protein, 50% lipid
(4) 30% protein, 50% lipid
100. Choose correct one w.r.t. composition of algae cell wall :-
(1) Cellulose + Galactans + Mannans + Minerals like CaCO_3
(2) Hemicellulose and pectin
(3) Cellulose + Hemicellulose + Pectin + Minerals like CaCO_2
(4) Suberin + Cellulose + Galactans + Silica
101. Cathepsin enzyme is helpful in digestion of tail of tadpole larva, it is secreted by :-
(1) Sphaerosome (2) Peroxisome
(3) Lysosome (4) Glyoxysome
102. The principle protein of Cilia and Flagella is :-
(1) Tubulin (2) Albumin
(3) Globulin (4) Fibrinogen
103. Function of centriole is :-
(1) Formation of spindle fibre
(2) Formation of Nucleolus
(3) Initiation of cell membrane
(4) Formation of cell plate
104. The site of synthesis and degradation of H_2O_2 in a plant cell is :-
(1) Lysosome
(2) Sphaerosome
(3) Peroxisome
(4) Microsome
105. Microtubules are present in all, except :-
(1) Flagella
(2) Spindle fibres
(3) Centrioles
(4) Microbodies
106. An elaborate network of filamentous and proteinaceous structures present in the cytoplasm of an eukaryotic cell is called :-
(1) Cytoskeleton
(2) Chromatin fibres
(3) Polysome
(4) Microsome



Identify P, Q, R and S in above given diagram. Which is showing ingestion of food material?

Option :-

- (1) P → Pinocytosis, Q → Phagosome, R → Phagocytosis, S → Pinosome
- (2) P → Phagocytosis, Q → Phagosome, R → Pinocytosis, S → Pinosome
- (3) P → Pinocytosis, Q → Pinosome, R → Phagocytosis, S → Phagosome
- (4) P → Pinosome, Q → Pinocytosis, R → Phagosome, S → Phagocytosis

108. Disc shaped structure present on the side of primary constriction meant for spindle fibre attachment are called :-

- (1) Secondary constriction
- (2) Centriole
- (3) Kinetochore
- (4) Diplosome

109. Nucleus was first observed by Robert Brown in :-

- (1) Maize root (2) *Acetabularia*
- (3) Orchid root (4) *Fritillaria*

110. Both cilium and flagellum of a Eukaryotic cell emerge from a structure which shows :-

- (1) 9 + 2 microtubule organisation
- (2) Cart wheel like organisation of microtubule
- (3) A pair of centrally located microtubule
- (4) Unit membrane bound structure

111. Nucleolus is the main site for the synthesis of :-

- (1) Ribosomal subunits
- (2) Proteins
- (3) Euchromatin
- (4) Heterochromatin

112. Centrosome is an organelle containing two cylindrical structure :-

- (1) Cisternae (2) Tubules
- (3) Centrioles (4) Centromere

113. Which of the following organelles is involved in the conversion of fat into carbohydrates?

- (1) Peroxisome
- (2) Lysosome
- (3) Glyoxysome
- (4) Mesosome

114. Non membranous organelle without DNA with duplicating nature is :-

- (1) Plasmid (2) Centriole
- (3) Ribosome (4) Plastid

115. Meristematic cells have

- (1) Primary cell wall
- (2) Secondary cell wall
- (3) Tertiary cell wall
- (4) All

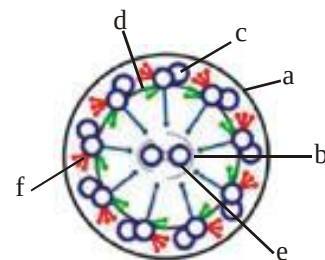
116. Glycosidation of proteins and lipids occur in _____.

- (1) Plastids (2) Vacuoles
- (3) Golgi apparatus (4) Lysosomes

117. Which of the following pair is incorrect?

- (1) Basal granule – centriole like structure
- (2) Axoneme – core of cilia
- (3) Radial spokes – connect two peripheral tubules
- (4) Centriole – 9 + 0

118. Identify the figure and find the correct option



- (1) a-cell membrane, e- interdoublet bridge, c- peripheral doublet, f-dynein arms
- (2) c-peripheral doublet e-linker, d-nexin, a-cell membrane
- (3) a-cell membrane, c-linker, d-dynein arms, f-radial spokes
- (4) b-central sheath, c- peripheral doublet, e-central singlet, f-dynein arm

119. Mitochondria and chloroplast are
(i) Semiautonomous organelle
(ii) Endosymbionts
(iii) Derive from pre-existing organelle
(iv) They do not have protein synthesis machinery

- (1) i & ii (2) i, ii & iii
(3) i, iii & iv (4) i, ii, iii & iv

120. All are functions of cytoskeleton except

- (1) Mobility of cell
(2) Repair of cell membrane
(3) Cell shape
(4) Support to organelles

121. _____ is a polymorphic organelle.

- (1) Golgi body (2) Lysosome
(3) ER (4) Ribosome

122. Water soluble pigments are found in

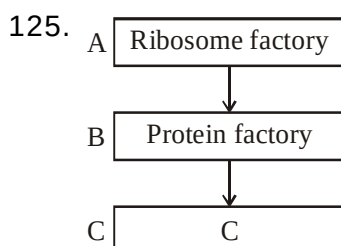
- (1) Chromoplast (2) Vacuole
(3) Mitochondria (4) All

123. Which statement is not appropriate?

- (1) Centrioles are surrounded by amorphous pericentriolar bodies
(2) Outer and inner nuclear membranes fuse to form nuclear pores at a number of places
(3) Numerous nucleoli present in cells actively carrying out protein synthesis
(4) Chromatin consist of DNA and RNA

124. Which of the following pair is correct?

- (1) histone - acidic amino acids
(2) non histone - basic amino acids
(3) centromeres - first site of constriction
(4) kinetochore - cylindrical structure



Identify A, B and C respectively?

- (1) A → Ribosome, B → Nucleolus, C → Protein
(2) A → Nucleolus, B → Lysosome, → Lipid
(3) A → Nucleus, B → Ribosome, C → Protein
(4) A → Nucleolus, B → Ribosome, C → Protein

126. Read the following statements. How many statements are correct ?

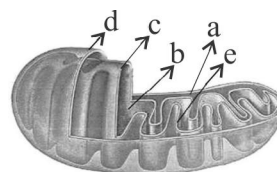
- (a) Basal granules have cart wheel like organization
(b) ER helps in synthesis of lipoproteins and glycogen
(c) RuBisCO is most abundant protein in biosphere
(d) The number of chloroplast varies from 1 in chlmydomonas to 20-40 per cell in the mesophyll.

- (1) three (2) two
(3) four (4) one

127. _____ is present in germinating seeds and concerned with oxidation of fats

- (1) Peroxisomes
(2) Glyoxysomes
(3) Sphaerosomes
(4) Lomasomes

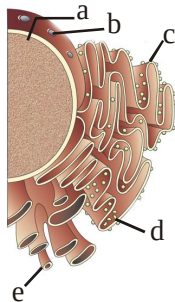
128. Identify the figure and find the correct match



- (1) a-inner membrane, b-crista, c-outer membrane, d-perimitochondrial space, e-matrix
(2) a-Intermembrane space, b-matrix, c-outer membrane, d-inner membrane, e-crista
(3) a -perimitochondrial space, b-matrix, c-inner membrane, d-outer membrane, e-crista
(4) a-crista, b-matrix, c-inner membrane, d-outer membrane, e-perimitochondrial space

129. Concentration of ions is higher in sap vacuole than cytoplasm because
- (1) ions diffuse from cytoplasm into vacuole
 - (2) tonoplast is more permeable for ions than other materials
 - (3) tonoplast has active transport system that pump ions from cytoplasm into sap vacuole
 - (4) interior side of tonoplast is impermeable to ions

130. Identify the figure and find the correct match



- (1) a -nucleus, b -nucleopore, c -RER, d -ribosomes, e-SER
 - (2) a -nucleus, b - nucleopore, c - SER, d - ribosomes, e-RER
 - (3) a-nucleolus, b -nucleopore, c -RER, d -ribosomes, e-SER
 - (4) a -nucleolus, b -nucleopore, c -SER, d -ribosomes, e-RER
131. Single membrane bound organelle is
- (1) ER
 - (2) Golgi body
 - (3) Lysosome
 - (4) All
132. Cell wall component hemicellulose synthesized by which cell organelle?
- (1) Endoplasmic reticulum
 - (2) Sphaerosome
 - (3) Golgibody
 - (4) Lysosome
133. Synthesis of r-RNA occurs in
- (1) nucleus
 - (2) nucleolus
 - (3) cytoplasm
 - (4) ribosome
134. Which structures are made up of microtubules?
- (1) Centrosome, nucleosome, centrioles
 - (2) Cilium, centriole, spindle fibre
 - (3) Centriole, spindle fibre, chromatin
 - (4) Cilium, flagellum, ergastic substance

135. Which of the following statement is not correct?

- (1) The thylakoid membrane contains photosynthetic pigments such as chlorophyll, carotenes and xanthophylls
 - (2) Lysosomes help in remodeling of cartilage and bones.
 - (3) Microfilaments are present in muscle fibers and associated with myosin protein
 - (4) None of these
136. Mendel conducted hybridisation experiments on garden peas from:
- (1) 1842-1847
 - (2) 1856-1863
 - (3) 1863-1873
 - (4) 1820-1831
137. The phenotypic ratio of a monohybrid cross in F_2 generation is:
- (1) 3 : 1
 - (2) 1 : 2 : 1
 - (3) 2 : 1 : 1
 - (4) 1 : 3 : 1
138. Which of the following trait of garden pea is not present on 4th chromosome?
- (1) Pod shape
 - (2) Flower position
 - (3) Flower colour
 - (4) Stem height
139. Which of the following genotype represents heterozygous condition?
- (1) AABB
 - (2) aaBB
 - (3) AaBb
 - (4) aabb
140. Who performed hybridisation experiments on garden pea before Mendel?
- (1) Kolreuter
 - (2) Darwin
 - (3) Hugo de Vries
 - (4) Correns
141. When two hybrids are crossed, the percentage of dominant is:
- (1) 75 %
 - (2) 25 %
 - (3) 50%
 - (4) 100%
142. Types of gametes produced by an individual heterozygous for three genes is :
- (1) 8
 - (2) 6
 - (3) 9
 - (4) 16
143. Type of gametes formed by the plant with genotype Aa bb CC Dd:
- (1) 4
 - (2) 8
 - (3) 16
 - (4) 32

144. How many zygotic combinations are possible between a cross $AA\ BB\ Cc\ Dd \times aa\ Bb\ Cc\ Dd$?
(1) 32 (2) 64 (3) 16 (4) 18
145. A dwarf pea plant is treated with gibberellin which makes it tall. This plant is crossed with dwarf pea plant. The phenotypic ratio in next generation shall be :
(1) 3 tall: 1 dwarf
(2) 50 % tall: 50 % dwarf
(3) All tall
(4) All dwarf
146. If an organism produces 32 types of gametes. Its genotype should be heterozygous for:
(1) 4 genes (2) 5 genes
(3) 8 genes (4) 2 genes
147. A pea plant heterozygous for seed shape and seed colour was selfed and total of 800 seeds were produced. The number of seeds with both homozygous recessive traits is:
(1) 50 (2) 250
(3) 200 (4) 100
148. Heterozygous tall (Tt) and dwarf (tt) pea plants are crossed. The ratio in the offsprings shall be :
(1) All tall
(2) All dwarf
(3) 50 % tall: 50 dwarf
(3) 3 tall: 1 dwarf
149. In a cross between two pea plants ($Rr\ Yy$ and $rryy$), the percentage of progeny with yellow seeds will be:
(1) 50% (2) 100%
(3) 75% (3) 25%
150. Proportion of offspring with genotype $aa\ Bb\ Cc\ Dd$ in a cross $Aa\ BB\ Cc\ DD \times aa\ bb\ Cc\ Dd$ will be :-
(1) $1/16$ (2) $4/16$
(3) $1/8$ (4) $8/16$
151. Sum total of genotypes and phenotypes in F_2 generation of a Mendelian trihybrid cross is
(1) 27 (2) 35 (3) 64 (4) 16
152. According to the concept of dominance, the modified allele is equivalent to the unmodified allele when:
(1) It produces normal/less efficient enzyme
(2) It produces the same phenotype/trait
(3) It results in the transformation of substrate
(4) More than one option is correct
153. From a cross $AA\ Bb \times aa\ Bb$, the genotypes $Aa\ BB : Aa\ Bb : Aa\ bb : aa\ bb$ are obtained in ratio
(1) $1 : 2 : 1 : 1$ (2) $0 : 3 : 1 : 0$
(3) $1 : 2 : 1 : 0$ (4) $1 : 1 : 1 : 0$
154. What is the probability of obtaining individuals with genotype $Pp\ qq\ Rr$, when a cross is made between the parents with genotype $Pp\ Qq\ Rr \times pp\ qq\ rr$? Also identify the type of cross.
(1) $\frac{1}{16}$, reciprocal cross
(2) $\frac{1}{8}$, test cross
(3) $\frac{1}{8}$, out cross
(4) $\frac{4}{64}$, test cross
155. When a heterozygous tall pea of F_1 generation upon self fertilization produces tall and dwarf phenotypes it proves the principle of :-
(1) Dominance
(2) Segregation
(3) Independent assortment
(4) Inheritance & purity of gametes
156. Back cross is a cross between :-
(1) $F_1 \times F_1$
(2) $F_1 \times$ Recessive
(3) $F_1 \times$ Dominant
(4) $F_1 \times$ any parent

157. A normal woman whose father was albino, marries an albino man, what proportion of normal and albino are expected among their offsprings :-
 (1) All normal
 (2) 2 normal : 1 Albino
 (3) All albino
 (4) 1 normal : 1 Albino
158. What is the ratio of a progeny of cross between F_1 heterozygous and recessive :-
 (1) 3 : 1
 (2) 1 : 2 : 1
 (3) 1 : 1
 (4) 2 : 1
159. In a cross between a pure tall plant with green pod and pure short plant with yellow pod how many short plants are produced in F_2 generation out of 16 :-
 (1) 1 (2) 3 (3) 4 (4) 9
160. If Mendel has chosen to study traits determined by linked genes he would not have discovered-
 (1) Law of segregation
 (2) Law of dominance
 (3) Law of independent assortment
 (4) Law of unit factor
161. On crossing blue & white flowered plants only blue coloured flowers produced, then on selfing the blue flowered plants, the offsprings would be:-
 (1) 72 : 24 (2) 40 : 60
 (3) 52 : 48 (4) 84 : 16
162. Pea plants were more suitable than cattle for Mendel's experiment because :-
 (1) There were no breeding records of cattle
 (2) Pea plants can be self-fertilised
 (3) Cattle are not easy to maintain
 (4) All pea plants have 2X chromosomes and fewer genetic traits
163. An individual with two identical members of a pair of genetic factors is called:-
 (1) Heteromorphic
 (2) Heterozygote
 (3) Homomorphic
 (4) Homozygous
164. In a cross 45 tall & 14 dwarf plants were obtained, genotype of parents would be :-
 (1) TT X TT (2) TT X Tt
 (3) Tt X Tt (4) TT X tt
165. When two hybrids $rrTt$ & $Rrtt$ are crossed, the phenotypic ratio of offspring will be :-
 (1) 3 : 1 (2) 1 : 1 : 1 : 1
 (3) 1 : 1 (4) 9 : 3 : 3 : 1
166. Marriages between close relatives should be avoided because it includes more :-
 (1) Recessive alleles to come together
 (2) Mutations
 (3) Multiple births
 (4) Blood group abnormalities
167. 1 : 1 : 1 : 1 ratio shows—
 (1) Monohybrid cross
 (2) Dihybrid cross
 (3) Back cross
 (4) Dihybrid test cross
168. In which type of inheritance the results are affected by reciprocal cross :-
 (1) Nuclear (2) Cytoplasmic
 (3) Blending (4) All the above
169. The word "Genetics" coined by :-
 (1) Mendel (2) Johannsen
 (3) Bateson (4) Morgan
170. If a heterozygous tall plant is crossed with a homozygous dwarf plant then what shall be the percentage of dwarf in offspring :-
 (1) 25% (2) 100%
 (3) 75% (4) 50%
171. A useful process for determining whether an individual is homozygous or heterozygous is -
 (1) Cross breeding (2) Self fertilisation
 (3) Back Crossing (4) Inbreeding
172. Blue eye colour in human is recessive to brown eye colour. The expected children of a marriage between a blue eyed woman and a brown eyed man who has a blue eyed mother will be
 (1) All black eyed
 (2) All blue eyed
 (3) All brown eyed
 (4) One blue and one brown eyed
173. Cross $AaBb \times AaBb$ yields $AaBB$; $AaBb$; $Aabb$: $aabb$ offspring in the ratio of :-
 (1) 0 : 3 : 1 : 1 (2) 1 : 2 : 1 : 0
 (3) 1 : 4 : 2 : 1 (4) 2 : 4 : 2 : 1

174. A tobacco plant heterozygous for albinism is self pollinated and 1200 seeds are subsequently germinated. How many seedlings would have the parental genotype :

- (1) 900 (2) 600
(3) 1200 (4) 300

175. The scientist who worked on garden pea before mendel was :-

- (1) Goss
(2) Kolreuter
(3) Bateson
(4) Maupertius

176. Read the following statements about hybridization technique of mendel and select the correct statement:-

- (A) Emasculation of both male and female parents is required.
(B) Emasculation of only male parent is required.
(C) Emasculation of female parent is required.
(D) Emasculation is not necessary.

- (1) A (2) B
(3) C (4) All correct

177. A cross is made between $AaBB \times AABb$ genotype individuals, then find out the correct option of probabilities of following offsprings

	A AABB	B AaBB	C AaBb	D aabb
(1)	1/4	1/4	1/4	0
(2)	1/8	1/8	4/16	0/16
(3)	1/16	2/16	2/8	1/16
(4)	2/8	2/8	4/8	0/16

178. Which of the following type of gamete will not be produced normally by genotype $AaBbCc$?

- (1) AbC (2) ABc
(3) aBC (4) aBb

179. The Punnett square shown below represents the pattern of inheritance in dihybrid cross, when round (R) is dominant over wrinkled (r) seeds and yellow (Y) is dominant over green (y) seeds.

	RY	Ry	rY	ry
RY	F	J	N	R
Ry	G	K	O	S
rY	H	L	P	T
ry	I	M	Q	U

Find out the odd one –

- (1) G (2) K (3) H (4) L

180. Which one of the following characters chosen by Mendel can express in heterozygous condition ?

- (1) Green seed colour
(2) Constricted pod shape
(3) Terminal flower position
(4) Green pod colour

